KTH ROYAL INSTITUTE
OF TECHNOLOGY

Degree Project in Decision and Control Systems

First cycle, 30 credits

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MARKUS MULVIHILL

Stockholm, Sweden, 2026

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MARKUS MULVIHILL

Master’s Programme, Information and Network Engineering, 120 credits
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Supervisors: A. Busy Supervisor, Another Busy Supervisor, Third Busy Supervisor
Examiner: Gerald Q. Maguire Jr.
School of Electrical Engineering and Computer Science
Host company: Företaget AB
Swedish title: Detta är den svenska översättnin gen av titeln
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Abstract

All theses at KTH are **required** to have an abstract in both *English* and *Swedish*.

Exchange students many want to include one or more abstracts in the language(s) used in their home institutions to avoid the need to write another thesis when returning to their home institution.

Keep in mind that most of your potential readers are only going to read your `title` and `abstract`. This is why it is important that the abstract give them enough information that they can decide is this document relevant to them or not. Otherwise the likely default choice is to ignore the rest of your document.
A abstract should stand on its own, i.e., no citations, cross references to the body of the document, acronyms must be spelled out, ...
Write this early and revise as necessary. This will help keep you focused on what you are trying to do.

Enter your abstract here!

Write an abstract that is about 250 and 350 words (1/2 A4-page) with the following components:

- What is the topic area? (optional) Introduces the subject area for the project.
- Short problem statement
- Why was this problem worth a Bachelor's/Master's thesis project? (*i.e.*, why is the problem both significant and of a suitable degree of difficulty for a Bachelor's/Master's thesis project? Why has no one else solved it yet?)
- How did you solve the problem? What was your method/insight?
- Results/Conclusions/Consequences/Impact: What are your key results/ conclusions? What will others do based upon your results? What can be done now that you have finished - that could not be done before your thesis project was completed?

The following are some notes about what can be included (in terms of LaTeX) in your abstract.

Choice of typeface with \textit, \textbf, and \texttt: *x*, **x**, and `x`.

Text superscripts and subscripts with \textsubscript and \textsuperscript: A_x and A^x.

Some symbols that you might find useful are available, such as: \textregistered, \texttrademark, and \textcopyright. For example, the copyright symbol: \textcopyright Maguire 2022 results in ©Maguire 2022. Additionally, here are some examples of text superscripts (which can be combined with some symbols): ^{99m}Tc, A*, A^{\textregistered}, and A\texttrademark resulting in ^{99m}Tc, A^*, $A^®$, and A™. Two examples of subscripts are: H\textsubscript{2}O and CO\textsubscript{2} which produce H_2O and CO_2.

Simple environments with begin and end: itemize and enumerate and within these \item.

The following commands can be used: \eg, \Eg, \ie, \Ie, \etc, and \etal: *e.g.*, *E.g.*, *i.e.*, *I.e.*, *etc.*, and *et al.*

The following commands for numbering with lower case Roman numerals: \first, \Second, \third, \fourth, \fifth, \sixth, \seventh, and \eighth: (i), (ii), (iii), (iv), (v), (vi), (vii), and $(viii)$. Note that the second case is set with a capital 'S' to avoid conflicts with the use second of as a unit in the `siunitx` package.

Equations using \(xxxx \) or \[xxxx \] can be used in the abstract. For example: $(C_5O_2H_8)_n$ or

$$\int_a^b x^2\,dx$$

Note that you **cannot** use an equation between dollar signs.

Even LaTeX comments can be handled, for example: % comment. Note that one can include percentages, such as: 51% or 51 %.

Keywords

Canvas Learning Management System, Docker containers, Performance tuning

Choosing good keywords can help others to locate your paper, thesis, dissertation, . . . and related work.

Choose the most specific keyword from those used in your domain, see for example: the ACM Computing Classification System (`https://www.acm.org/publications/computing-classification-system/how-to-use`), the IEEE Taxonomy (`https://www.ieee.org/publications/services/the`

saurus-thank-you.html), PhySH (Physics Subject Headings) (https://physh.aps.org/), ... or keyword selection tools such as the National Library of Medicine's Medical Subject Headings (MeSH) (https://www.nlm.nih.gov/mesh/authors.html) or Google's Keyword Tool (https://keywordtool.io/)

Formatting the keywords:

- The first letter of a keyword should be set with a capital letter and proper names should be capitalized as usual.
- Spell out acronyms and abbreviations.
- Avoid "stop words" - as they generally carry little or no information.
- List your keywords separated by commas (",").

Since you should have both English and Swedish keywords - you might think of ordering them in corresponding order (*i.e.*, so that the n^{th} word in each list correspond) - this makes it easier to mechanically find matching keywords.

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Enter your Swedish abstract or summary here!

Alla avhandlingar vid KTH **måste ha** ett abstrakt på både *engelska* och *svenska*.
Om du skriver din avhandling på svenska ska detta göras först (och placera det som det första abstraktet) - och du bör revidera det vid behov.

If you are writing your thesis in English, you can leave this until the draft version that goes to your opponent for the written opposition. In this way you can provide the English and Swedish abstract/summary information that can be used in the announcement for your oral presentation.
If you are writing your thesis in English, then this section can be a summary targeted at a more general reader. However, if you are writing your thesis in Swedish, then the reverse is true – your abstract should be for your target audience, while an English summary can be written targeted at a more general audience.
This means that the English abstract and Swedish sammnfattning or Swedish abstract and English summary need not be literal translations of each other.

Do not use the \glspl{} command in an abstract that is not in English, as my programs do not know how to generate plurals in other languages. Instead you will need to spell these terms out or give the proper plural form. In fact, it is a good idea not to use the glossary commands at all in an abstract/summary in a language other than the language used in the `acronyms.tex file` - since the glossary package does **not** support use of more than one language.

The abstract in the language used for the thesis should be the first abstract, while the Summary/Sammanfattning in the other language can follow

Nyckelord

Canvas Lärplattform, Dockerbehållare, Prestandajustering

Nyckelord som beskriver innehållet i uppsatsen eller rapporten

Acknowledgments

Författarnas tack

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I detta kapitel kan du ev nämna något om din bakgrund om det påverkar rapporten på något sätt. Har du t ex inte möjlighet att skriva perfekt svenska för att du är nyanländ till landet kan det vara på sin plats att nämna detta här. OBS, detta får dock inte vara en ursäkt för att lämna in en rapport med undermåligt språk, undermålig grammatik och stavning (t ex får fel som en automatisk stavningskontroll och grammatikkontroll kan upptäcka inte förekomma)
En dualism som måste hanteras i hela rapporten och projektet

I would like to thank xxxx for having yyyy. Or in the case of two authors: We would like to thank xxxx for having yyyy.

Stockholm, March 2026
Markus Mulvihill

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Listings

If you have listings in your thesis. If not, then remove this preface page.

List of acronyms and abbreviations

This document is incomplete. The external file associated with the glossary 'acronym' (which should be called `thesis.acr`) hasn't been created.

The file `thesis.acn` doesn't exist. This most likely means you haven't used `\makeglossaries` or you have used `\nofiles`. If this is just a draft version of the document, you can suppress this message using the `nomissingglstext` package option.

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The list of acronyms and abbreviations should be in alphabetical order based on the spelling of the acronym or abbreviation.

Chapter 1

Introduction

Patients in the intensive care unit suffering from critical illnesses (e.g. acute myocardial infarction, sepsis, stroke) have been widely documented to undergo hyperglycaemia, even those without a prior diagnosis of diabetes [1]. Acute stress observed in critical illnesses often induces physiological changes, as the secretion of cortisol and catecholamines results in an inflammatory response associated with insulin-resistance [2] [3] [4] [5] [6] [7]. While there is inconclusive evidence pertaining to the relationship of stress hyperglycaemia and adverse outcomes, the onset of mortality is far greater in patients with no reported history of hyperglycaemia, in contrast to patients with diabetes [1] [3]. Sustaining glucose concentration under 6.106 mmol/L through insulin infusion has demonstrated a reduction of mortality and morbidity, compared to less-intensive methods of glycaemic control [8], at the subsequent cost of an increase of reported incidences of hypoglycaemia [9]. Further review of intensive insulin therapy and tight glucose control has identified discrepancies in the administration of glycaemic control, mostly corresponding to the risk of moderate to severe hypoglycaemia and glycaemic variability in a critically ill patient is equally detrimental as stress hyperglycaemia, with an increase of mortality and health complications observed [9] [10] [11]. Clinicians aim to maintain blood glucose in a range of 6 mmol/L to 10 mmol/L to account for adverse outcomes in hyperglycaemia and hypoglycaemia. Traditional intensive insulin therapy struggles to fully capture the dynamic fluctuations of blood glucose in critically ill patients. Thus, more personalized model-based methods are required to achieve tight glycaemic control.

1.1 Background

The implementation of state feedback in model-based tight glucose control methods relies on the formulation of derived state estimators of insulin-glucose dynamics. Mechanistic insulin-glucose models have been developed with an extensive physiological foundation, and have been validated for gluco-regulation in clinical settings [12] [13] [14] [15] [16]. The insulin-glucose models derived are nonlinear, which requires more advanced state estimation methods to properly estimate the insulin-glucose dynamics in patients, as well as prediction into the future. The patient parameters used in such estimators must reflect glucose-insulin dynamics observed physiologically to accurately model gluco-regulation. Mechanistic insulin-glucose models published have traditionally assigned patient parameters as population constants, backed by parameter sensitivity analysis studies. There are data-driven methods to approximate some patient parameters as shown in [14] [16], but a lack of estimators for a subset of patient-specific parameters has been implemented along with the nonlinear state estimation of insulin-glucose dynamics. As a result, this current methodology of estimating insulin-glucose dynamics is not designed to be patient-specific.

1.2 Problem

1.2.1 Original problem and definition

There currently exists a gap in research for further fine-tuning of mechanistic insulin-glucose models. While there is evidence shown in sensitivity studies that certain patient-specific parameters do not have much influence in the insulin-glucose dynamics [12] [15] [16], inadequate assigning of the remaining parameters can lead to inadequate glycaemic control in critically ill patients.

1.2.2 Scientific and engineering issues

System identification can be employed to model the behavior of dynamical systems. Experience of statistical modeling within system identification is often required to construct the assumptions of the system parameters. In the case of insulin-glucose dynamics, there are several unknown patient-specific parameters, and each has its respective role in modeling gluco-regulation. As a result, identifying key patient-specific parameters and the methods to model them are not intuitive, and poses an ongoing challenge.

1.3 Purpose

Mechanistic insulin-glucose models are of great benefit to critically ill patients and clinicians, as modeling gluco-regulation allows for necessary glycaemic adjustments to avoid the onset of adverse outcomes. In practice, clinicians only have the ability to base insulin sensitivity on blood glucose measurements, and thus, state estimators models fill in the gap as a tool for insulin-glucose dynamics estimation. Improvements in the estimation of insulin-glucose dynamics can better capture the evolution of the health of the patient in the intensive care unit to prevent mortality and morbidity.

1.4 Goals

The overall goal of this project is the estimation of patient-specific parameters in mechanistic insulin-glucose models to compare to current methods. This has been divided into the following sub-goals:

1. Estimate insulin-glucose dynamics using a nonlinear state estimator
2. Determine the subset of patient-specific parameters of interest to model
3. Formulate estimators for chosen patient-specific parameters

1.5 Research Methodology

The modeling of the insulin-glucose dynamics found in critically ill patients will be implemented on a virtual patient cohort. Using virtual patients allows the researcher to manually control the patient-specific parameters. The control of the experiment will be insulin-glucose state estimators without any parameter estimation (i.e. assigning parameters as population constants), while the treatment group incorporates parameter estimation with the state estimators. To evaluate the performance of the study, the control group and the treatment group will estimate the insulin-glucose dynamics of unseen virtual patients, and fitting metrics of the state estimators are utilized to draw inferences about the performance of the parameter estimation.

1.6 Delimitations

This study will not consist of any real patient data from the intensive care unit. Using virtual patients is important for maintaining internal validity in the experimental process, as it verifies the observed outcome of the experiment is due to the manipulation of patient-specific parameters and not from uncontrolled variables. This study will build the foundation of the modeling of insulin-glucose dynamics, and future work can replicate the study with preprocessed real patient data.

1.7 Structure of the thesis

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Chapter 2

Background

The Intensive Control Insulin-Nutrition-Glucose model presented by Lin et al. [14] characterizes the insulin-glucose dynamics in critically ill patients. The model is constructed as a fusion of two existing models validated for patients in the intensive care unit [12] [13], containing the following ensemble of ordinary differential equations

$$\dot{BG} = -p_G BG(t) - S_I BG(t)\frac{Q(t)}{1+\alpha_G Q(t)} + \frac{P(t) + EGP_B - CNS}{V_G} \tag{2.1}$$

$$\dot{Q} = n_I(I(t) - Q(t)) - n_C\frac{Q(t)}{1+\alpha_G Q(t)} \tag{2.2}$$

$$\dot{I} = -n_K I(t) - n_L\frac{I(t)}{1+\alpha_I I(t)} - n_I(I(t) - Q(t)) + \frac{u_{ex}(t) + (1-x_L)u_{en}(t)}{V_I} \tag{2.3}$$

$$\dot{P1} = -d_1 P1 + D(t) \tag{2.4}$$

$$\dot{P2} = -min(d_2 P2, P_{max}) + d_1 P1 \tag{2.5}$$

$$P(t) = -min(d_2 P2, P_{max}) + PN(t) \tag{2.6}$$

$$u_{en}(t) = k_1 e^{-\frac{k_2}{k_3}I(t)}, \quad \text{when C-peptide data is not available.} \tag{2.7}$$

Equation 2.1 is the independent insulin glucose removal, where BG is the total blood glucose concentration (mmol/L). The whole-body insulin sensitivity S_I gives insight to the metabolic condition of the patient and is a time-varying dynamic fitted hourly to the patient. The insulin pharamacodynamics are shown in equations 2.2 and 2.3, with Q and I being the interstitial insulin concentration (mU/L) and the plasma insulin concentration (mU/L) respectively. The exogenous insulin input is represented as $u_{ex}(t)$ (mU/min), and the endogenous insulin production $u_{en}(t)$ (mU/min) can be measured from C-Peptide, but if measurements are not feasible, the kinetics can modeled as such in equation 2.7. The gastric absorption of glucose is defined in equations 2.4 to 2.6. The dynamics $P1$ and $P2$ describe the glucose in the stomach and the gut (mmol), with $P(t)$ representing the glucose appearance in the blood (mmol/min). Any enteral food intake given to the patient is accounted for by $D(t)$ (mmol/min), and additional parenteral dextrose is described by $PN(t)$ (mmol/min).

The Intensive Control Insulin-Nutrition-Glucose model also includes several fixed parameters detailed in table 2.1. The majority of the patient model parameters are assigned as general population constants observed from clinical trials, and are validated in relevant insulin-glucose control parameter sensitivity studies [15] [16] [17]. The identification of the remaining constant parameters were estimated with a data-driven approach (i.e. model identifiability) by fitting the patient parameters to the observed dynamics in 173 patients, and by assessing the impact of prediction errors. Estimating the endogenous glucose removal p_G and the suppression of endogenous glucose produce EGP_b constants involved assigning all other patient parameters and the insulin sensitivity to constants, then starting a grid search that covers $p_G = 0.001, 0.002, \ldots, 0.1(\text{min}^{-1})$ and $EGP_b = 0.0, 0.1, \ldots, 3.5$ (mmol/min). The joint p_G and EGP_b value that achieved the lowest fitting and prediction error for blood glucose in equation 2.1 was chosen, ultimately that was $p_G = 0.006(\text{min}^{-1})$ and $EGP_b = 1.16$, while both falling in the ranges observed physiologically. The glucose distribution rate between the stomach and the gut n_I, and cellular insulin clearance rate in the interstitium were estimated similarly by fitting to Q in equation 2.2, a grid search covers a range $n_I = n_C = 0.0001, \ldots, 0.02(\text{min}^{-1})$, with a value of $n_I = 0.003$ resulting in the best predictive performance.

While the proposed clinically-validated model for patients in the intensive care unit has shown to capture long-term insulin-glucose dynamics, in practice, typically it is only possible for clinicians to take blood glucose measurements, consequently, excluding measurements for the dynamics seen in equations 2.2-2.5. Thus, leveraging estimators is a viable solution to estimate the insulin-glucose dynamics over time to account for the lack of intuition into the dynamics that can not be measured.

Table 2.1: Constant Parameters in the Intensive Control Insulin-Nutrition-Glucose Model.

Constant Parameter	**Notation**	**Value**
Patient endogenous glucose removal	p_G (min^{-1})	0.006
Saturation of insulin-stimulated glucose removal	α_G (L/mU)	0.0154
Suppression of endogenous glucose production	EGP_b (mmol/min)	1.16
Central nervous system uptake	CNS (mmol/L)	0.3
Glucose distribution volume	V_G (L)	13.3
Glucose distribution rate between the stomach and the gut	n_I (min^{-1})	0.003
Cellular insulin clearance rate in the interstitium	n_C (min^{-1})	$= n_I$
Kidney clearance rate of insulin from plasma	n_K (min^{-1})	0.0542
Liver clearance rate of insulin from plasma	n_L (min^{-1})	0.01578
Saturation of plasma insulin disappearance	α_I (L/mU)	0.0017
Insulin distribution volume	V_I (L)	3.15
Fraction of endogenous insulin hepatic extraction	x_L	0.67
Maximal glucose out flux from the gut	P_{max} (mmol/min)	6.11
Base endogenous insulin production rate	k_1 (mU/min)	45.7
Transport rates between the stomach and the gut	d_1 (min^{-1})	0.0347
	d_2 (min^{-1})	0.0069
Exponential suppression constants	k_2	1.5
	k_3	1000

Parameters shaded in gray are derived from fitting methods, as the rest are assigned as population constants.

2.1 State Estimation

A probabilistic state estimator can be modeled in the following state space representation

$$\begin{aligned} x_k &\sim p(x_k \mid x_{k-1}, u_{k-1}) \\ y_k &\sim p(y_k \mid x_k), \end{aligned} \tag{2.8}$$

where $p(x_k, u_k|x_{k-1})$ and $p(y_k|x_k)$ are the discrete-time dynamic and measurement models, $x_k \in \mathbb{R}^n$ are the states, $u_k \in \mathbb{R}^p$ are the inputs, and $y_k \in \mathbb{R}^m$ are the measurements. The objective is to ultimately derive the point estimate $\hat{x_k}$ of the state, given all the measurements up to the current sample $p(x_k \mid y_{1:k})$. In the case of assuming the state estimator is linear and Gaussian,

$$\begin{aligned} x_k &= A_{k-1}x_{k-1} + B_{k-1}u_{k-1} + q_{k-1} \\ y_k &= Hx_k + r_k, \end{aligned} \tag{2.9}$$

where $q_{k-1} \sim N(0, Q_{k-1})$ is the process noise, $r_k \sim N(0, R_k)$ is the measurement noise, we can rewrite the equation 2.8 as the following

$$\begin{aligned} p(x_k \mid x_{k-1}, u_{k-1}) &= N(x_k \mid A_{k-1}x_{k-1} + B_{k-1}u_{k-1}, Q_{k-1}) \\ p(y_k \mid x_k) &= N(y_k \mid H_k x_k, R_k). \end{aligned} \tag{2.10}$$

The closed-form solution of $p(x_k \mid y_{1:k})$ for linear and Gaussian dynamic and measurement functions is known as the Kalman filter [18], with two key parts: the *prediction step* and the *update step*.

1. *Prediction step*

$$\begin{aligned} m_k^- &= A_{k-1}x_{k-1} + B_{k-1}u_{k-1} \\ P_k^- &= A_{k-1}P_{k-1}A_{k-1}^T + Q_{k-1}. \end{aligned} \tag{2.11}$$

2. *Update step*

$$\begin{aligned} v_k &= y_k - H_k m_k^- \\ S_k &= H_k P_k^- H_k^T + R_k \\ K_k &= P_k^- H_k^T S_k^{-1} \\ m_k &= m_k^- + K_k v_K \\ P_k &= P_k^- - K_k S_k K_k^T. \end{aligned} \tag{2.12}$$

The Kalman Filter is recursive, as the filter starts with an initial point estimate of the mean m_0^- and the covariance P_0^-, and iterates over discrete time steps k. While the Kalman filter is a great starting point for an estimator, as the assumptions of linearity and Gaussianity often hold in many contexts, it is clear when analyzing the equations 2.1 to 2.5, the ordinary differential equations are not linear, hence, the estimators of choice must have the capability to properly handle nonlinearities in the dynamic function.

2.1.1 Extended Kalman filter

The extended Kalman filter expands upon the linearity condition to allow for nonlinear models. The structure of the extended Kalman filter remains the same as the equations 2.13 and 2.14, but changes are seen in the matrices of the dynamic and measurement models $f(\cdot)$ and $h(\cdot)$. Assuming the same Gaussianity condition as the Kalman filter, the prediction and update steps [18] can be defined as

1. *Prediction step*

$$\begin{aligned} m_k^- &= f(m_{k-1}, u_{k-1}) \\ P_k^- &= F_x(m_{k-1}) P_{k-1} F_x^T(m_{k-1}) + Q_{k-1}, \end{aligned} \tag{2.13}$$

2. *Update step*

$$\begin{aligned} v_k &= y_k - h(m_k^-) \\ S_k &= H_x(m_k^-) P_k^- H_x^T(m_k^-) + R_k \\ K_k &= P_k^- H_k^T(m_k^-) S_k^{-1} \\ m_k &= m_k^- + K_k v_K \\ P_k &= P_k^- - K_k S_k K_k^T, \end{aligned} \tag{2.14}$$

where $F_x(m_k^-)$ and $H_x(m_k^-)$ are the Jacobian matrices of $f(\cdot)$ and $h(\cdot)$ derived from the Taylor series Gaussian approximation of a linear transformation, and are written as

$$\begin{aligned} [F_x(m)]_{j,j'} &= \left.\frac{\partial f_j(x)}{\partial x_{j'}}\right|_{x=m} \\ [H_x(m)]_{j,j'} &= \left.\frac{\partial h_j(x)}{\partial x_{j'}}\right|_{x=m}. \end{aligned} \tag{2.15}$$

2.1.2 Bootstrap particle filter

When revisiting the probabilistic state estimators in section 2.1, to analytically compute the point estimate of the states $\hat{x}$, given all observations up to time step K, one must evaluate the expected value of the posterior probability

$$\hat{x} = E[g(x) \mid y_{1:K}] = \sum_x g(x) p(x \mid y_{1:K}), \tag{2.16}$$

where $g(\cdot)$ is an arbitrary distribution $\mathbb{R}^n \rightarrow \mathbb{R}^m$. Such an integral is often difficult to compute in closed-form due to the dimensionality of x, but by formulating a Monte Carlo approximation by drawing N many random samples from the target distribution $x^i \sim g(x) \quad i = 1, \ldots, N$, the approximation of the point estimate of the states will converge to the true point estimate according to the Law of Large Numbers and the Central Limit Theorem for a large N [19], and the approximation of the state point estimate can be rewritten as

$$\hat{x} \approx \frac{1}{N} \sum_{i=1}^{N} g(x^i) p(x^i \mid y_{1:K}). \tag{2.17}$$

The conditional probability $p(x^i \mid y_{1:K})$ is often unknown, and can not be evaluated, but according to Bayes' rule, it can be rewritten as

$$\begin{aligned} p(x^i \mid y_{1:K}) &= \frac{p(y_{1:K} \mid x^i)p(x^i)}{\frac{1}{N}\sum_{j=1}^{N} p(y_{1:K} \mid x^j)p(x^j)} \\ &= \frac{p(y_{1:K} \mid x^i)p(x^i)}{\pi(x^i \mid y_{1:K})}, \end{aligned} \tag{2.18}$$

where $\pi(x^i \mid y_{1:K})$ is defined as the importance mass, and is a Monte Carlo approximation of the conditional probability of $p(x^i \mid y_{1:K})$, with the condition being that the mass is non-zero whenever $p(x^i \mid y_{1:K})$ is non-zero, which results in a nonuniform distribution. Sampling from a nonuniform distribution will result in each particle having a corresponding weight $w^{*(i)}$, and a normalized weight w^i

$$\begin{aligned} w^{*(i)} &= \frac{p(y_{1:K} \mid x^i)p(x^i)}{\pi(x^i \mid y_{1:K})} \\ w^i &= \frac{w^{*(i)}}{\sum_{j=1}^{N} w^{*(j)}}, \end{aligned} \tag{2.19}$$

with the approximation of the point estimate being rewritten as

$$\hat{x} \approx \sum_{i=1}^{N} w^i g(x^i). \tag{2.20}$$

The bootstrap particle filter is formulated on the basis of Markov Chain Monte Carlo approximations, where the filtering distributions are shown in equation 2.8, and is run sequentially like the Kalman filter, with the importance mass being assigned as the dynamic model $p(x_k \mid x_{k-1})$, which results in the following algorithm [18] [19]

1. Sample N particles from the dynamic function

$$x_k^i = p(x_k \mid x_{k-1}^i, u_{k-1}) \quad i = 1, \dots, N \tag{2.21}$$

2. Update weights from the measurement

$$\begin{aligned} w_k^{*(i)} &\propto p(y_k \mid x_k^i) \quad i = 1, \dots, N \\ w_k^i &= \frac{w_k^{*(i)}}{\sum_{j=1}^{N} w_k^{*(j)}} \quad i = 1, \dots, N \end{aligned} \tag{2.22}$$

3. Resample particles from a distribution proportional to the weights.

2.2 Related Work

As previously mentioned, the whole-body insulin sensitivity S_I dynamically fluctuates over time depending on the patient's metabolic condition, as a result, naively assigning S_I as a constant parameter can hinder the performance of the estimators. A nonlinear and nonconvex integral-based parameter identification method has been introduced that can approximate patient-specific parameters [15]. By linearly interpolating between the blood glucose measurements at times t_0 and t_1,

$$\widetilde{BG(t)} = \frac{BG(t_1) - BG(t_0)}{t_1 - t_0} t, \tag{2.23}$$

the equation 2.1 be rearranged as such

$$\begin{aligned}
\int_{t_0}^{t_1} \dot{BG}\, dt &= \begin{pmatrix} -p_G \int_{t_0}^{t_1} \widetilde{BG(t)}\, dt - S_I \int_{t_0}^{t_1} \widetilde{BG(t)} \frac{Q(t)}{1 + \alpha_G Q(t)}\, dt \\ + \frac{1}{V_G} \int_{t_0}^{t_1} P(t) + EGP_b - CNS\, dt \end{pmatrix} \\
BG(t_1) - BG(t_0) &= \begin{pmatrix} -p_G \int_{t_0}^{t_1} \widetilde{BG(t)}\, dt - S_I \int_{t_0}^{t_1} \widetilde{BG(t)} \frac{Q(t)}{1 + \alpha_G Q(t)}\, dt \\ + \frac{1}{V_G} \int_{t_0}^{t_1} P(t) + EGP_b - CNS\, dt \end{pmatrix} \\
\underbrace{\int_{t_0}^{t_1} \widetilde{BG(t)} \frac{Q(t)}{1 + \alpha_G Q(t)}\, dt}_{A(t_0, t_1)}\ S_I &= \underbrace{\begin{pmatrix} BG(t_0) - BG(t_1) - p_G \int_{t_0}^{t_1} \widetilde{BG(t)}\, dt \\ + \frac{1}{V_G} \int_{t_0}^{t_1} (P(t) + EGP_b - CNS)\, dt \end{pmatrix}}_{b(t_0, t_1)}
\end{aligned} \tag{2.24}$$

It is often the case with longer measurement periods $(t_1 - t_0)$, that identifying multiple S_I values within the time period may be necessary. For instance, if every 120 mins glucose measurements are taken from the patient, but every 60 mins a patient's insulin sensitivity should be identified $S_I \in \mathbb{R}^2$, both insulin sensitivity parameters at 60 mins S_I^{60} and 120 mins S_I^{120} can be derived as a least squares solution from the system of linear equations

$$\begin{bmatrix} A(t_0, t_0+30) & 0 \\ A(t_0+30, t_0+60) & 0 \\ 0 & A(t_0+60, t_0+90) \\ 0 & A(t_0+90, t_0+120) \end{bmatrix} \begin{bmatrix} S_I^{60} \\ S_I^{120} \end{bmatrix} = \begin{bmatrix} b(t_0, t_0+30) \\ b(t_0+30, t_0+60) \\ b(t_0+60, t_0+90) \\ b(t_0+90, t_0+120) \end{bmatrix}. \tag{2.25}$$

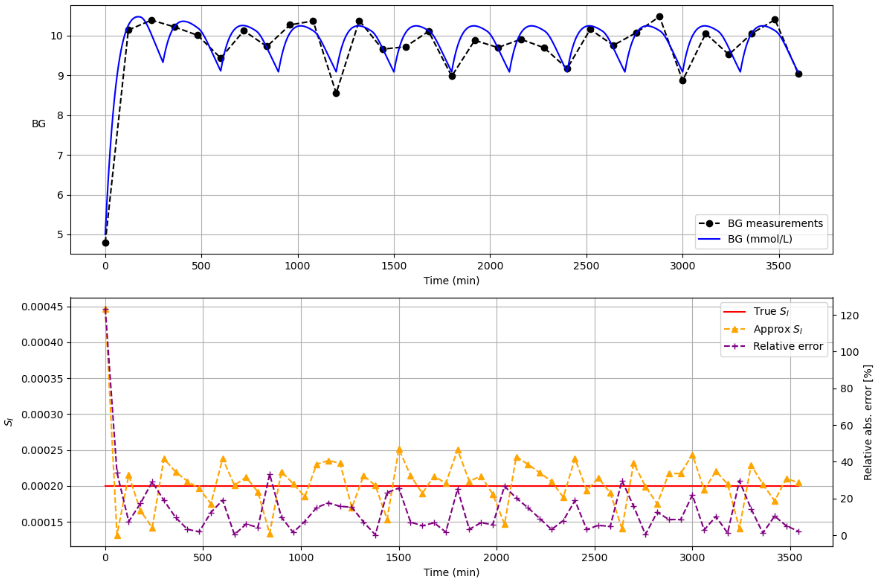


Figure 2.1: Integral-based Insulin Sensitivity Identification on Patient Model*.

2.3 Summary

*Initial conditions of the patient model are $BG = 5.0$ (mmol/L), $I = 15.0$ (mU/L), $Q = 15.0$ (mU/L), $P1 = P2 = 0.0$ (mmol) due to no external nutrition given. Measurements are sampled with additive Gaussian noise $r_k \sim N(0, 0.25^2)$.

Chapter 3

Method or Methods

Metod eller Metodval

This chapter is about Engineering-related content, Methodologies and Methods. Use a self-explaining title.
The contents and structure of this chapter will change with your choice of methodology and methods.

Describe the engineering-related contents (preferably with models) and the research methodology and methods that are used in the degree project.
Give a theoretical description of the scientific or engineering methodology you are going to use and why have you chosen this method. What other methods did you consider and why did you reject them.
In this chapter, you describe what engineering-related and scientific skills you are going to apply, such as modeling, analyzing, developing, and evaluating engineering-related and scientific content. The choice of these methods should be appropriate for the problem . Additionally, you should be consciousness of aspects relating to society and ethics (if applicable). The choices should also reflect your goals and what you (or someone else) should be able to do as a result of your solution - which could not be done well before you started.

The purpose of this chapter is to provide an overview of the research method used in this thesis. Section 3.1 describes the research process. Section 3.2 details the research paradigm. Section 3.3 focuses on the data collection techniques used for this research. Section 3.4 describes the experimental design. Section 3.5 explains the techniques used to evaluate the reliability and validity of the data collected. Section 3.6 describes the method used for the data analysis. Finally, Section 3.7 describes the framework selected to evaluate xxx.

Vilka vetenskaplig eller ingenjörs-metodik ska du använda och varför har du valt den här metoden. Vilka andra metoder gjorde du övervägde du och varför du avvisar dem. Vad är dina mål? (Vad ska du kunna göra som ett resultat av din lösning - vilken inte kan göras i god tid innan du började) Vad du ska göra? Hur? Varför? Till exempel, om du har implementerat en artefakt vad gjorde du och varför? Hur kommer du utvärdera den. Syftet med detta kapitel är att ge en översikt över forsknings metod som används i denna avhandling. Avsnitt 3.1 beskriver forskningsprocessen. Avsnitt 3.2 beskriver forskningsparadigmen detaljerat. Avsnitt 3.3 fokuserar på datainsamlingstekniker som används för denna forskning. Avsnitt 3.4 beskriver experimentell design. Avsnitt 3.5 förklarar de tekniker som används för att utvärdera tillförlitligheten och giltigheten av de insamlade uppgifterna. Avsnitt 3.6 beskriver den metod som används för dataanalysen. Slutligen, Avsnitt 3.7 beskriver ramverket som valts för att utvärdera xxx.
Ofta kan man koppla ett antal följdfrågor till undersökningsfrågan och problemlösningen t ex
(1) Vilken process skall användas för konstruktion av lösningen och vilken process skall kopplas till denna för att svara på undersökningsfrågan?
(2) Hur och vilket resultat (storheter) skall presenteras både för att redovisa svar på undersökningsfrågan (resultatkapitlet i denna rapport) och redovisa resultat av problemlösningen (prototypen, ofta dokument som bilagor men vilka dokument och varför?).
(3) Vilken teori/teknik skall väljas och användas både för undersökningen (taxonomi, matematik, grafer, storheter mm) och problemlösning (UML, UseCases, Java mm) och varför?
(4) Vad behöver du som student leverera för att uppnå hög kvaliet (minimikrav) eller mycket hög kvalitet på examensarbetet?
(5) Frågorna kopplar till de följande underkapitlen.
(6) Resonemanget bygger på att studenter på hing-programmet ofta skall konstruera något åt problemägaren och att man till detta måste koppla en intressant ingenjörsfråga. Det finns hela tiden en dualism mellan dessa aspekter i exjobbet.

3.1 Research Process

Undersökningsrocess och utvecklingsprocess

Figure 3.1 shows the steps conducted in order to carry out this research.

Figur 3.1 visar de steg som utförs för att genomföra
Beskriv, gärna med ett aktivitetsdiagram (UML?), din undersökningsprocess och utvecklingsprocess. Du måste koppla ihop det akademiska intresset (undersökningsprocess) med ursprungsproblemet (utvecklingsprocess) denna forskning.
Aktivitetsdiagram från t ex UML-standard

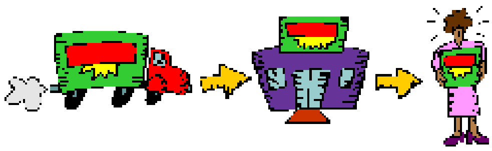


Figure 3.1: Research Process

Example of using customize item labels.

Some steps in the process:

Step 1 plan experiment,

Step 2 conduct experiment,

Step 3 analyze data from the experiment, and

Step 4 discuss the results of the analysis.

Forskningsprocessen

3.2 Research Paradigm

Undersökningsparadigm
Exempelvis
Positivistisk (vad/hur fungerar det?) kvalitativ fallstudie med en deduktivt (förbestämd) vald ansats och ett induktivt(efterhand uppstår dataområden och data) insamlade av data och erfarenheter.

3.3 Data Collection

Datainsamling
(Detta bör också visa att du är medveten om de sociala och etiska frågor som kan vara relevanta för dina data insamlingsmetod.)

This should also show that you are aware of the social and ethical concerns that might be relevant to your data collection method.

3.3.1 Sampling

Stickprovsundersökning

3.3.2 Sample Size

Provstorleken

3.3.3 Target Population

Målgruppen

3.4 Experimental design and Planned Measurements

Experimentdesign/Mätuppställning

3.4.1 Test environment/test bed/model

Describe everything that someone else would need to reproduce your test environment/test bed/model/... .

Testmiljö/testbädd/modell
Beskriv allt att någon annan skulle behöva återskapa din testmiljö / testbädd / modell / ...

3.4.2 Hardware/Software to be used

Hårdvara / programvara som ska användas

3.5 Assessing reliability and validity of the data collected

Bedömning av validitet och reliabilitet hos använda metoder och insamlade data

3.5.1 Validity of method

Giltigheten av metoder
Har dina metoder gett dig de rätta svaren och lösningarna? Var metoderna korrekta?

How will you know if your results are valid?

Remember that validity is about the *accuracy* of a measurement while reliability is about the *consistency* of the measurement values under the same conditions (*i.e.,* repeatability).

3.5.2 Reliability of method

Tillförlitlighet av för metoder
Hur bra är dina metoder, finns det bättre metoder? Hur kan du förbättra dem?

How will you know if your results are reliable?

3.5.3 Data validity

Giltigheten av uppgifter
Hur vet du om dina resultat är giltiga? Är ditt resultat rättvisande?

3.5.4 Reliability of data

Tillförlitlighet av data
Hur vet du om dina resultat är tillförlitliga? Hur bra är dina resultat?

3.6 Planned Data Analysis

Metod för analys av data

3.6.1 Data Analysis Technique

Dataanalysteknik

3.6.2 Software Tools

Mjukvaruverktyg

3.7 Evaluation framework

Utvärdering och ramverk
Metod för utvärdering, jämförelse mm. Kopplar till kapitel 5.

3.8 System documentation

Systemdokumentation
Med vilka dokument och hur skall en konstruerad prototyp dokumenteras? Detta blir ofta bilagor till rapporten och det som problemägaren till det ursprungliga problemet (industrin) ofta vill ha. Bland dessa bilagor återfinns ofta, och enligt någon angiven standard, kravdokument, arkitekturdokument, designdokumnet, implementationsdokument, driftsdokument, testprotokoll mm.

If this is going to be a complete document consider putting it in as an appendix, then just put the highlights here.

Chapter 4

What you did

Choose your own chapter title to describe this

[Vad gjorde du? Hur gick det till? – Välj lämplig rubrik ("Genomförande", "Konstruktion", "Utveckling" eller annat]

What have you done? How did you do it? What design decisions did you make? How did what you did help you to meet your goals?

Vad du har gjort? Hur gjorde du det? Vilka designval gjorde du? Hur kom det du hjälpte dig att uppnå dina mål?

4.1 Hardware/Software design . . . /Model/Simulation model & parameters/. . .

Hårdvara / Mjukvarudesign ... / modell / Simuleringsmodell och parametrar / . . .

Figure 4.1 shows a simple icon for a home page. The time to access this page when served will be quantified in a series of experiments. The configurations that have been tested in the test bed are listed in Table 4.1. In 7.0 % of cases there was an error indicating xxxxx.

Figur 4.1 visar en enkel ikon för en hemsida. Tiden för att få tillgång till den här sidan när den laddas kommer att kvantifieras i en serie experiment. De konfigurationer som har testats i provbänk listas ini tabell 4.1.
Vad du har gjort? Hur gjorde du det? Vilka designval gjorde du?

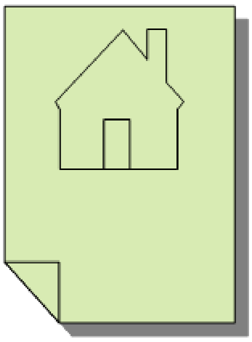


Figure 4.1: Homepage icon

Table 4.1: Configurations tested

Configuration	Description
1	Simple test with one server
2	Simple test with one server

Testade konfigurationer

4.2 Implementation . . . /Modeling/Simulation/. . .

Implementering . . . / modellering / simulering / . . .

Two commonly used simulators are:

Mininet This simulator uses traffic control (`tc`) to simulate network devices connected by links with specific bandwidth, packet loss rates, qdisc methods, etc.

ns-2 or ns-3 simulator These simulators are very useful for simulating wireless communication links between moving devices. You can specific the mobility patterns of the nodes.

4.2.1 Some examples of coding

This section is simply to show some example of how you can include code in your thesis - this is not a section you would have in your thesis.

Det här avsnittet är helt enkelt för att visa ett exempel på hur du kan inkludera kod i ditt examensarbete - det här är inte ett avsnitt du skulle ha i ditt examensarbete.

Listing 4.1 shows an example of a simple program written in C code.

Listing 4.1: Hello world in C code

```
int main() {
printf("hello,␣world");
return 0;
}
```

In contrast, Listing 4.2 is an example of code in Python to get a list of all of the programs at KTH.

Listing 4.2: Using a python program to access the KTH API to get all of the programs at KTH

```
KOPPSbaseUrl = 'https://www.kth.se'

def v1_get_programmes():
    global Verbose_Flag
    #
    # Use the KOPPS API to get the data
    # note that this returns XML
    url = "{0}/api/kopps/v1/programme".format(KOPPSbaseUrl)
    if Verbose_Flag:
        print("url:␣" + url)
    #
    r = requests.get(url)
    if Verbose_Flag:
        print("result␣of␣getting␣v1␣programme:␣{}".format(r.text)
    #
    if r.status_code == requests.codes.ok:
        return r.text           # simply return the XML
    #
    return None
```

4.2.2 Some examples of figures in tikz

This section is simply to show some example of how you can draw your own figures for in your thesis - this is not a section you would have in your thesis.

Det här avsnittet är helt enkelt för att visa ett exempel på hur du kan rita dina egna figurer i ditt examensarbete – det här är inte ett avsnitt du skulle ha i ditt examensarbete.

These figures are just some examples to show that you can draw your own figures for in your thesis. This has two advantages: (i) you do not have to worry about copyrights – as these are your own figures and (ii) the text is now readable and not simply a picture of text – so screen readers can read the figure's contents to someone who is listening to the contents of your thesis.

4.2.2.1 Azure's Form Recognizer

Figure 4.2 shows the processing of key-value extraction from a PDF document using Azure's Form Recognizer.

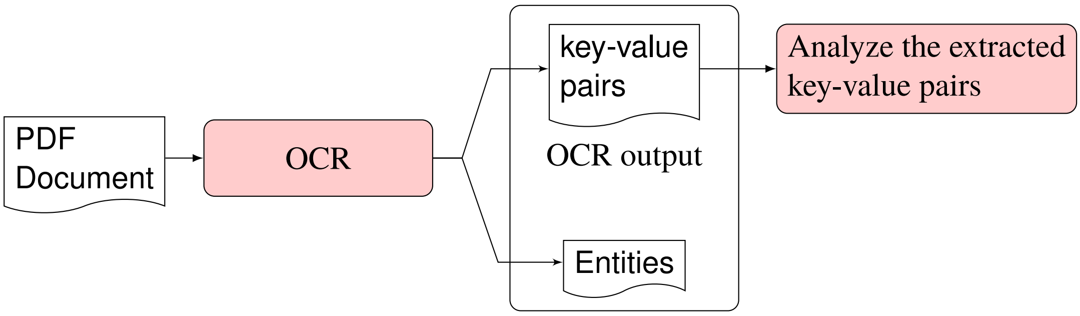


Figure 4.2: The processing of key-value extraction from a PDF document using Azure's Form Recognizer

4.2.2.2 Hyper-V with Containers

Figure 4.3 shows how Hyper-V deals with containers.

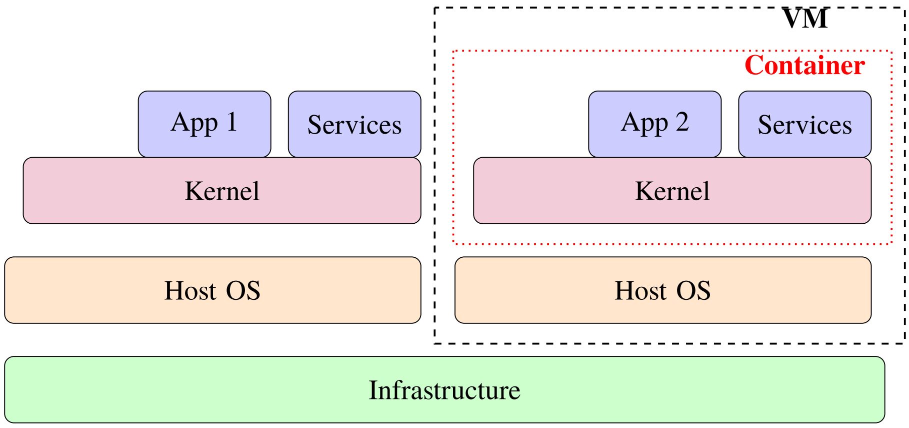
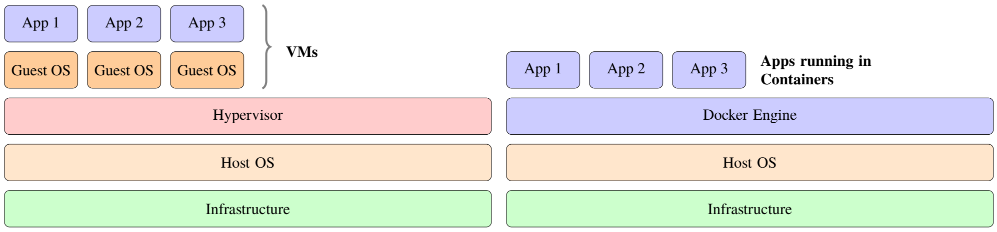


Figure 4.3: Hyper-V with containers

4.2.2.3 VM versus Containers

Figure 4.4 shows a comparison of virtual machines (VMs) versus containers.



(a) VM (b) Containers

Figure 4.4: Virtual machines (VMs) versus Containers

Chapter 5

Results and Analysis

svensk: Resultat och Analys

Sometimes this is split into two chapters.
Keep in mind: How you are going to evaluate what you have done?
What are your metrics?
Analysis of your data and proposed solution
Does this meet the goals which you had when you started?

In this chapter, we present the results and discuss them.

I detta kapitel presenterar vi resultaten och diskutera dem.
Ibland delas detta upp i två kapitel.
Hur du ska utvärdera vad du har gjort? Vad är din statistik?
Analys av data och föreslagen lösning
Innebär detta att uppfyllelse av de mål som du hade när du började?

5.1 Major results

Huvudsakliga resultat

Some statistics of the delay measurements are shown in Table 5.1. The delay has been computed from the time the GET request is received until the response is sent.

Lite statistik av fördröjningsmätningarna visas i Tabell 5.1. Förseningen har beräknats från den tidpunkt då begäran GET tas emot fram till svaret skickas.

Table 5.1: Delay measurement statistics

Configuration	**Average delay (ns)**	**Median delay (ns)**
1	467.35	450.10
2	1687.5	901.23

Table 5.2 shows the measurement of round trip times from four hosts to and from a server.

Table 5.2: Result for the ping measurements of RTT for 4 hosts

	host to server RTT in ms			
Host	min	avg	max	mdev
h1	5.625	5.625	5.625	0.0
h2	2.909	2.909	1.909	0.0
h3	5.007	5.007	5.007	0.0
h4	2.308	2.308	2.308	0.0

Fördröj mätstatistik

Konfiguration | Genomsnittlig fördröjning (ns) | Median fördröjning (ns)

Figure 5.1 shows an example of the performance as measured in the experiments.

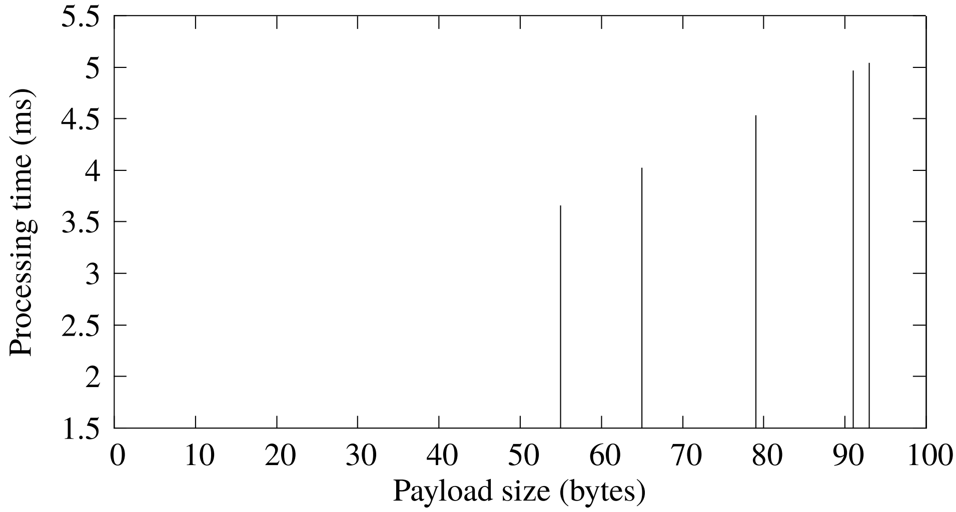


Figure 5.1: Processing time vs. payload length

Given these measurements, we can calculate our processing bit rate as the inverse of the time it takes to process an additional byte divided by 8 bits per byte:

$$\text{bit rate} = \frac{1}{\frac{\text{time}_{\text{byte}}}{8}} = 20.03 \quad kb/s$$

Table 5.3 shows another table in which some values have been set in bold (using \B) to emphasize them. Note how the `S` formatting has been modified so that it considers the weight of the characters and this is able to decimal align even these hold faced numbers with the numbers in the column above them.

Table 5.3: Median values of sandwich attributes

	sites	
Attribute	**A**	**B**
price (in SEK)	36.5	71.3
protean (g)	97.2	100.0
salt (mg)	9.7	9.3
Average customer rating in %	**82.2**	**89.9**

5.2 Reliability Analysis

Analys av tillförlitlighet
Tillförlitlighet i metod och data

5.3 Validity Analysis

Analys av validitet
Validitet i metod och data

Chapter 6

Discussion

Diskussion
Förbättringsförslag?

This can be a separate chapter or a section in the previous chapter.

Chapter 7

Conclusions and Future work

Slutsats och framtida arbete

Add text to introduce the subsections of this chapter.

7.1 Conclusions

Slutsatser

Describe the conclusions (reflect on the whole introduction given in Chapter 1).

Discuss the positive effects and the drawbacks.
Describe the evaluation of the results of the degree project.
Did you meet your goals?
What insights have you gained?
What suggestions can you give to others working in this area?
If you had it to do again, what would you have done differently?

Uppfyllde du dina mål?
Vilka insikter har du fått?
Vilka förslag kan du ge till andra som arbetar inom detta område? Om du skulle göra detta igen, vad skulle du ha gjort annorlunda?

7.2 Limitations

Begränsande faktorer
Vad gjorde du som begränsade dina ansträngningar? Vilka är begränsningarna i dina resultat?

What did you find that limited your efforts? What are the limitations of your results?

7.3 Future work

Vad du har kvar ogjort?
Vad är nästa självklara saker som ska göras?
Vad tips kan du ge till nästa person som kommer att följa upp på ditt arbete?

Describe valid future work that you or someone else could or should do. Consider: What you have left undone? What are the next obvious things to be done? What hints can you give to the next person who is going to follow up on your work?

Due to the breadth of the problem, only some of the initial goals have been met. In these section we will focus on some of the remaining issues that should be addressed in future work. ...

7.3.1 What has been left undone?

The prototype does not address the third requirment, *i.e.,* a yearly unavailability of less than 3 minutes, this remains an open problem. ...

7.3.1.1 Cost analysis

Example of a missing component

The current prototype works, but the performance from a cost perspective makes this an impractical solution. Future work must reduce the cost of this solution, to do so a cost analysis needs to first be done. ...

7.3.1.2 Security

Example of a missing component

A future research effort is needed to address the security holes that results from using a self-signed certificate. Page filling text mass. Page filling text mass. ...

7.3.2 Next obvious things to be done

In particular, the author of this thesis wishes to point out xxxxxx remains as a problem to be solved. Solving this problem is the next thing that should be done. ...

7.4 Reflections

Reflektioner

Vilka är de relevanta ekonomiska, sociala, miljömässiga och etiska aspekter av ditt arbete?

What are the relevant economic, social, environmental, and ethical aspects of your work?

One of the most important results is the reduction in the amount of energy required to process each packet while at the same time reducing the time required to process each packet.

The thesis contributes to the United Nations (UN) Sustainable Development Goals (SDGs) numbers 1 and 9 by xxxx.

In the references, let Zotero or other tool fill this in for you. I suggest an extended version of the IEEE style, to include URLs, DOIs, ISBNs, etc., to make it easier for your reader to find them. This will make life easier for your opponents and examiner.
IEEE Editorial Style Manual: https://www.ieee.org/content/dam/ieee-org/ieee/web/org/conferences/style_references_manual.pdf

Låt Zotero eller annat verktyg fylla i det här för dig. Jag föreslår en utökad version av IEEE stil - att inkludera webbadresser, DOI, ISBN etc. - för att göra det lättare för läsaren att hitta dem. Detta kommer att göra livet lättare för dina opponenter och examinator.

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If you do not have an appendix, do not include the \cleardoublepage command below, otherwise the last page number in the meta data will be one too large.

Appendix A

Something Extra

svensk: Extra Material som Bilaga

A.1 Just for testing KTH colors

You have selected to optimize for print output

- Primary color
  - kth-blue
  - kth-blue80
- Secondary colors
  - kth-lightblue
  - kth-lightred
  - kth-lightred80
  - kth-lightgreen
  - kth-coolgray
  - kth-coolgray80

black

Appendix B

Main equations

This appendix gives some examples of equations that are used throughout this thesis.

B.1 A simple example

The following example is adapted from Figure 1 of the documentation for the package nomencl (https://ctan.org/pkg/nomencl).

$$a = \frac{N}{A} \tag{B.1}$$

The equation $\sigma = ma$ follows easily from Equation (B.1).

B.2 An even simpler example

The formula for the diameter of a circle is shown in Equation (B.2) area of a circle is shown in eq. (B.3).

$$D_{circle} = 2\pi r \tag{B.2}$$

$$A_{circle} = \pi r^2 \tag{B.3}$$

Some more text that refers to (B.3).

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Enter your abstract here!

Write an abstract that is about 250 and 350 words (1/2 A4-page) with the following components:

- What is the topic area? (optional) Introduces the subject area for the project.
- Short problem statement
- Why was this problem worth a Bachelor's/Master's thesis project? (*i.e.*, why is the problem both significant and of a suitable degree of difficulty for a Bachelor's/Master's thesis project? Why has no one else solved it yet?)

- How did you solve the problem? What was your method/insight?
- Results/Conclusions/Consequences/Impact: What are your key results/ conclusions? What will others do based upon your results? What can be done now that you have finished - that could not be done before your thesis project was completed?

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"Keywords[eng]": €€€€
Canvas Learning Management System, Docker containers, Performance tuning €€€€,
"Abstract[swe]": €€€€

Enter your Swedish abstract or summary here!

Alla avhandlingar vid KTH **måste ha** ett abstrakt på både *engelska* och *svenska*.
Om du skriver din avhandling på svenska ska detta göras först (och placera det som det första abstraktet) - och du bör revidera det vid behov.

If you are writing your thesis in English, you can leave this until the draft version that goes to your opponent for the written opposition. In this way you can provide the English and Swedish abstract/summary information that can be used in the announcement for your oral presentation.
If you are writing your thesis in English, then this section can be a summary targeted at a more general reader. However, if you are writing your thesis in Swedish, then the reverse is true – your abstract should be for your target audience, while an English summary can be written targeted at a more general audience.
This means that the English abstract and Swedish sammnfattning or Swedish abstract and English summary need not be literal translations of each other.

Do not use the \glspl{} command in an abstract that is not in English, as my programs do not know how to generate plurals in other languages. Instead you will need to spell these terms out or give the proper plural form. In fact, it is a good idea not to use the glossary commands at all in an abstract/summary in a language other than the language used in the `acronyms.tex file` - since the glossary package does **not** support use of more than one language.

The abstract in the language used for the thesis should be the first abstract, while the Summary/Sammanfattning in the other language can follow

€€€€,
"Keywords[swe]": €€€€
Canvas Lärplattform, Dockerbehållare, Prestandajustering €€€€,
}

acronyms.tex

```
%%% Local Variables:
%%% mode: latex
%%% TeX-master: t
%%% End:
% The following command is used with glossaries-extra
\setabbreviationstyle[acronym]{long-short}
% The form of the entries in this file is \newacronym{label}{acronym}{phrase}
%                                       or \newacronym[options]{label}{acronym}{phrase}
% see "User Manual for glossaries.sty" for the  details about the options, one example is shown below
% note the specification of the long form plural in the line below
\newacronym[longplural={Debugging Information Entities}]{DIE}{DIE}{Debugging Information Entity}
%
% The following example also uses options
\newacronym[shortplural={OSes}, firstplural={operating systems (OSes)}]{OS}{OS}{operating system}

% note the use of a non-breaking dash in long text for the following acronym
\newacronym{IQL}{IQL}{Independent Q-Learning}  % <-- regular dash

\newacronym{KTH}{KTH}{KTH Royal Institute of Technology}

\newacronym{LAN}{LAN}{Local Area Network}
\newacronym{VM}{VM}{virtual machine}
% note the use of a non-breaking dash in the following acronym
\newacronym{WiFi}{Wi-Fi}{Wireless Fidelity}

\newacronym{WLAN}{WLAN}{Wireless Local Area Network}
\newacronym{UN}{UN}{United Nations}
\newacronym{SDG}{SDG}{Sustainable Development Goal}
```